

3. Lists, Indexing and IDs

Notice how in our last part, we wanted to have a variable for EVERY product in our program.

- Making a separate variable for every product is going to get messy so fast; LUCKILY, we can just group them in a *list*.

A LIST isn't anything special, it's literally just a list of values.

```
list = ["banana", "apple", "grape", "orange", 1, 4, 6, 7]
```

This is how you make a list `[]`, notice how it's not just words; a list can have words and numbers, or any combination of types.

- Now that I have this list of values, how do I get something out of it? I want to get out "banana", or "grape".

We get out the word "banana" by grabbing our list; and SPECIFYING a position in that list our value is in.

- You call it by doing `list[#]`. But, lists actually start at position 0. "banana" is at position 0; "apple" is at position 1.

`list[0]` is the SAME as banana; If I printed out `list[0]` it would print banana.

```
list = ["banana", "apple", "grape", "orange", 1, 4, 6, 7]

print(list[0])
```

For lists, the same way we can grab stuff out, we can actually ADD or REMOVE.

```
list.pop(0) OR lists.remove("banana")
```

This removes the value in position 0 from the list, pop allows you to do positions; since lists can have duplicate values.

```
lists.append("blueberry") OR lists.insert(1, "blueberry")
```

This ADDS the value blueberry, the command APPEND adds it to the end. And insert allows you to chose which position to add it into.

So, Now we can just have a **LIST**, with all of our products. But the issue is, *we'd have to make a WHOLE new list.*

- Or, we can turn our list into something called a **DICTIONARY**

Dictionary, is something that uses a key value pair. Meaning you **SEARCH** up a value, with a **UNIQUE** key value.

- Think of a book, if you've ever been in a library, **books may have the same name, but they'd have something called an ISBN** which is **UNIQUE** for each book. You use that ISBN to look stuff up.

A **dictionary** is done like below.

```
my_dictionary = {"something_Unique" : something_Not_Necessarily_Unique}
```

When we did lists a second ago, we **LOOKED** up a **value** via the **POSITION**.

- Here we look up values via our **UNIQUE VALUE**, called a **KEY** (on the left, *doesn't have to be a word, it can be a number too.*)

```
my_products = {"Banana" : 10.00, "Apple" : 10.00}
```

```
print(my_products["Banana"])
```

OK, *but we need to be able to add/remove* from the dictionary too. **It's simpler than list.**

To **ADD**, we do `my_products["blueberry"] = 15`

- It's **ADDING** because strawberry **doesn't exist**; if it **DID** exist it'd update that value.

To **DELETE**, we do `del my_products["blueberry"]`

Now we have a **NICE** way to save our data; this is actually **the standard way to save data.**

In the next part of the lab we'll cover something called an **IF STATEMENT**. *This is the basis of why you would even BOTHER programming. It allows us to make our program DO different things AUTOMATICALLY.*